

AI Core Specification

Project NOVA Cognitive Subsystems, Engines, and Planning Loops

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Revision History

Version	Date	Author	Summary of Changes
1.0	2026-06-28	Antigravity	Consolidate AI Kernel, Planner, Context, Memory, Voice, Vision, Execution, Observation, and Tool Manager specifications.

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Purpose & Scope

This specification details the structural design, boundaries, inputs, and outputs of Project NOVA's cognitive subsystems. It coordinates the data interactions between planning, observation, context gathering, and tool executions.

AI Kernel Core Orchestration

Purpose

Provide central orchestration to coordinate every AI engine and loop execution.

Responsibilities

- Receives natural user intent statements from input adapters.
- Invokes the Context Engine to assemble active system metrics.
- Delegates data sets to the Planner Engine to compile a plan sequence.
- Coordinates memory recalls and audits permissions with the Tool Manager.
- Returns synthesized status outputs back to response adapters.

Constraint: The AI Kernel does not execute system-level automation directly. It operates exclusively as a cognitive coordinator.

Planner Engine Specification

Purpose

Convert natural user intentions and active context states into structured, executable plans.

Interface inputs

- **Intent:** Structured classification of the user Goal.
- **Context:** Unified active system snapshot.
- **Memory:** Context histories and user configuration weights.

Interface outputs

- An ordered plan containing a sequence of target tasks.
- Alternative fallback plans to handle failure scenarios.

Constraint: The Planner Engine is passive. It compiles actions but never executes them directly.

Context Engine Specification

Purpose

Compile the active OS, browser, and user conversation status into a unified context JSON payload for the Planner.

Collected Metrics

- Active application name and process identifier.
- Focused window geometry and active coordinates.
- Clipboard string contents.
- Session conversation histories.
- Environment configurations.

Memory Engine Specification

Purpose

Store, retrieve, and rank text structures and user configs to maintain continuity across sessions.

Memory Layers

- **Short-Term Memory:** Conversation history within the active Session.

- **Long-Term Memory:** Historical facts and preferences persisted across sessions.
- **Workspace Memory:** Configured layout variables and active folder parameters.
- **Semantic Memory:** Embeddings maps representing system tools and commands.

Tool Manager & Adapter Boundary

Purpose

Register, expose, and route execution tasks to concrete tools.

Responsibilities

- Maintains a registry of available tools (Git, Playwright, OCR, local compilers).
- Exposes tools to the Planner for matching.
- Inspects permission settings before running any tool wrappers.

Execution & Observation Subsystem

Execution Engine

- Receives approved plans from the AI Kernel.
- Integrates with adapters to run clicks, keystrokes, and shell files.
- Operates strictly as an execution worker (never generates plans).
- Emits result updates to the Observation Engine.

Observation & Reflection

- **Observation:** Verifies the visual outcomes of clicks and key emulations.
- **Reflection:** Detects plan execution failures, flags errors, and sends recovery suggestions back to the Planner.

Voice Engine Specification

Purpose

Expose speech interfaces to communicate with the platform.

Responsibilities

- Performs configurable wake-word and push-to-talk microphone captures.
- Transcribes speech stream frames into text blocks (Speech-to-Text).

- Synthesizes response text into voice streams (Text-to-Speech).

Vision Engine Specification

Purpose

Acquire display screens and understand layout structures.

Responsibilities

- Captures screenshot frames via the Desktop Capability.
- Extracts layout-preserving text characters (OCR).
- Identifies GUI targets (buttons, forms, icons) and outputs structured boundaries.

Constraint: The Vision Engine is read-only. It parses screenshots but does not trigger mouse movements or actions.

AI Core Milestones

Implementation of the AI Core package follows this strict sequence:

11. **AI Kernel:** Coordinates loop flows.
12. **Planner:** Decomposes intentions.
13. **Memory:** Persists semantic tables.
14. **Context:** Aggregates window details.
15. **Voice:** Exposes audio input/output.
16. **Vision:** Translates screen pixels.
17. **Tool Manager:** Configures adapters.
18. **Execution:** Emulates keyboard/mouse inputs.
19. **Observation:** Verifies display outputs.
20. **Reflection:** Recovers from execution errors.